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Involvement of the fossil fuel industry in climate change mitigation efforts: Impacts on climate policymaking and on the pace of action

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Abstract

The climate crisis took the centre stage in 2021 with IPCC releasing its Sixth Assessment Report in July amid the COVID-19 pandemic and the lockdown, followed by the 2021 UN Climate Change Conference in Glasgow (COP26), held in the month of November. 2021 was a year of contradictions too, on the climate front. On the one hand, Unilever went coal-free, and on the other, the Glasgow climate pact was adopted after a last-minute intervention by India to water down language on "phasing out" coal to merely "phasing down", insinuating that coal will most definitely be around in the near future. Therefore, to understand humanity's stance regarding climate change, we will be examining how the fossil fuel industry has dealt with the climate crisis over time, starting from 1965 when anthropogenic climate change was for the very first time recognized by the world as a serious problem that poses a threat to humanity if left unchecked. The effects of the fossil fuel industry's participation in climate change mitigation talks on policymaking and the pace of the urgently needed change will be the key themes of this article.

INTRODUCTION

Anthropogenic climate change was acknowledged on a large scale for the first time in 1965 when the U.S. President's Science Advisory Committee issued a warning that human activities are negatively altering the climate. The UN mandated the Intergovernmental Panel on Climate Change (IPCC) in 1988 to clarify this issue through the evidence already available and to arrive at a scientific consensus that would assist policymakers, which is when the real action began to take place. The Rio Earth Summit in 1992 saw the adoption of the United Nations Framework Convention on Climate Change (UNFCCC), which while not legally obligatory on the parties still necessitates regular meetings between the ratifying nations (known as the Conference of the Parties, or COP).

UNFCCC officially spearheaded international climate policymaking. At COP3 in Japan in 1997, the Kyoto Protocol—the first legally binding climate treaty was enacted. The treaty finally went into effect in 2005, following several issues. All these significant actions culminated in the so-called "climate change regime" [1, 2, 3], which saw the entire world join the conversation about climate change as one, cohesive entity. In the years to follow, many summits, pacts, as well as several actors and fora devoted to combating climate change, would be seen to take birth.

In one such initiative, the UN's NAZCA platform [4], more than 2000 enterprises with a combined revenue of $US 36.6 trillion and 16 of the 20 largest banks in the world made public climate commitments. We Mean Business, a leading business climate action network, also brings together close to 700 businesses and investors with a combined market capitalization of over US$ 8 trillion [5]. The network also provides its members with vouchers to "reduce greenhouse gas emissions, build climate resilience, and unlock climate finance" so they can sustain themselves in a low-carbon or carbon-neutral economy [6]. 83 of these 700 companies, including Apple and IKEA, have made it clear that they are committed to achieving the target of switching entirely to renewable energy sources by 2030 [7].

The Paris Agreement of 2015 marked a turning point in the history of efforts to reduce anthropogenic climate change; 196 nations signed what experts hailed as the most important global climate accord in history. The Paris Agreement pursues efforts to keep the increase in global temperature below 1.5°C and to limit it to 2°C maximum.

Resistance from the Fossil Fuel Industry: Strategies Adopted

The oil and gas sector had adopted a very clear public stance against climate policy at a time when it was only beginning to gain traction. In 1989 i.e., one year after the IPCC was established, the top oil, gas, and coal companies got together to form the Global Climate Coalition (GCC), which supported "climate scepticism" [8, 9]. It represented roughly 40 worldwide corporations and industry associations from industries including paper and automobiles.

As time passed, GCC's foundation weakened [10]. Shell and BP departed the GCC in 1996, but Exxon, Mobil, and Texaco (now Chevron) remained until 2000 [11, 12, 16-20]. BP formally acknowledged anthropogenic climate change in 1997 after leaving the GCC [13], and in 1998 it surpassed Suntech as the largest vertically integrated solar enterprise [14]. Several reputable, older business associations supported oil and gas-related companies as well, although they refrained from direct climate denial as the GCC did. Before the 2nd Conference of the Parties (COP2), which was held in Geneva in 1993, the International Chamber of Commerce (ICC) assisted in

coordinating business positions, and the International Petroleum Environmental Conservation Association (now simply known as IPIECA) broadened its scope from a focus on the UN Environment Program (UNEP) to include the UNFCCC [15].

Hove *et al*., 2002, classified the approaches taken by oil and gas companies to combat climate change as (1) placing priority on the business consequences while standing against the claims of anthropogenic climate change; (2) avoiding accountability; and (3) prioritising the need for a business process modification keeping climate change mitigation in mind, while limiting the negative effect in terms of business consequences [21].

The businesses that were against emission regulations asserted that policies aimed at reducing global warming posed a threat to their industry and the economy. They argued that when they attempted to invest and switch to energy-saving alternatives and renewables, they incurred incorrigible losses. As a result, they are now cautious to avoid making the same error again. They have pointed out in the past that other industries, such as agriculture, are the true sources of greenhouse gas emissions. They stand by the belief that modern technology will soon offer the best climate change answers. [21]

Companies opting to "wait and see" believe that since the science of climate change is still debatable, any action or attitude taken at this time would be rash and without much consideration or investigation. Another defence is that since fossil fuels will continue to be used for a very long time, it will be difficult to eradicate them entirely in the near future. They cite the most recent data and studies, some of which indicate that coal, oil, and particularly natural gas (because of its clean nature) will rule the economy for many years to come. They believe that rather than taking action in a hurry, now is the moment to gather information and deepen our understanding of climate change. [21]

The "Proactive" lobby acknowledges that climate change poses a hazard to the world and, consequently, to their enterprises. They consider climate change to be one of the greatest problems facing humanity today, one that, if not addressed quickly, will have a significant impact on the economy. Prior to BP's withdrawal from the GCC in 1996 and, more crucially, BP CEO John Browne's historic address at Stanford University in May 1997, BP was a member of the "battle against emission limitations" lobby. It was a pleasant surprise when BP made the decision to implement a new (proactive) approach before Kyoto.

Involvement of Oil and Gas Majors in Climate Talks

By 2016, the majority of fossil fuel companies had acknowledged climate change as a significant problem. What sparked this change?

In contrast to what businesses and individuals predicted, it might be linked to the rapid expansion of renewable energy sources. The unit costs of renewable technologies had decreased, which contributed to this rise. A number of public policies promoting renewables had also begun to emerge, including G7 and G20 initiatives [22]. The renewables sector is predicted to grow by 650% before 2030, and with the appropriate regulations, that number may go much further [23]. According to McKinsey & Company, global investments in clean energy surpassed those in fossil fuels in the first quarter of 2015, and the total amount invested was close to US$ 300 billion [24].

The increasing momentum of climate-related public and private policy, the adoption of ambitious climate change mitigation goals by states and communities, and the establishment of a regulatory floor by the Paris Agreement may have all contributed to the acceptance of the existence of anthropogenic climate change. "As carbon policy strengthens, the oil and gas majors will confront increased regulatory burden and are likely to incur increasing costs," says Paul McConnell of industry research producer Wood Mackenzie. Hence, by 2016, the majority of fossil fuel companies had begun to take essential measures to prevent the decline of their energy-intensive operations in a potentially carbon-neutral future.

Things again started to get bitter between the green lobby and the fossil fuel lobby in the spring of 2016 when Ecuador and Venezuela sought a conflict-of-interest policy to limit the presence of fossil fuel representatives at UNFCCC events because most of them have budgets much larger than those of small states. The US, UK, and Australia initially rejected the concept, calling it "unclear" and "unfeasible" [25]. Then, just before COP22, the non-governmental organisation Corporate Accountability International (CAI) took matters into its own hands and amassed more than 500,000 signatures for a petition seeking to remove fossil fuel lobbyists from UN Climate Change negotiations [26, 27]. The United States assessed the circumstances, changed its stance, and accepted the petition. Another Organisation that has attempted to raise awareness of the aforementioned issues is Influence Map [28].

More than 500 fossil fuel lobbyists attended the most recent COP26 session, setting a record high and surpassing the number of representatives sent by any one nation. The green lobby was outraged, and some climate activists demanded that the sector completely refrain from participating in such climate-related activities.

Oil and Gas Companies and Climate Change Governance: Existing Strategies, Policies, and Low-Carbon Visions

There has been an effort to cut emissions in two different methods, one of which, roughly speaking, works in favour of the fossil fuel industry, and the other works against it.

The fossil fuel divestment movement, which urges investors to withdraw money from companies that deal in oil, gas, and coal, and the idea of the stranded asset, which advocates leaving the majority of fossil fuels in the ground to prevent climate change, aim to marginalise the role of oil and gas in a low-carbon future.

The divestment movement gained traction after the Rockefeller Family Fund made the decision to stop investing in fossil fuels. Given that it is controlled by the successors of the founders of Standard Oil, the forerunner to oil tycoons like ExxonMobil and Chevron, it was an unusual decision. The Fund believes that there is no sensible rationale for firms to continue to explore new sources of hydrocarbons [29]. Shell defended the fund's divestment, calling it a "logical approach" [30]. The divestment campaign currently has 500 or so groups involved, controlling more than US$ 3.4 trillion in assets.

The idea of "stranded assets" has been adopted by people who are giving up fossil fuels. Many Governments [31], international organisations [32, 33, 34] and financial institutions [35, 36] have been seen to support the idea. Only a portion of the world's fossil fuel reserves is thought to be recoverable if we are to stay under the 2°C temperature increase threshold. Due to the financial risks involved, which include the possibility that some of the reserves upon which the oil and gas companies and their shareholders are valued could become stranded, resistance from these companies and their shareholders to this idea is common.

The demand for oil and gas companies to develop strategies to maintain their relevance and competitiveness in the years after the Paris Agreement grows intensely as the years go

by. Shareholders are asking fossil fuel companies to "stress-test" their business plans against pacts like the Paris Agreement in addition to providing a detailed explanation of how they plan to respond to an increase in the number of climate change mitigation policies. Also, some businesses are being pressured to disclose any stranded assets and unlink executive pay from the growth in reserves [37]. Some corporations have managed to sidestep such demands [38], but a great many of them have begun articulating their own plans [39, 40, 41]. Efficiency in energy use, carbon price, and carbon capture and storage are three important aspects of these plans (CCS).

Both CCS and carbon pricing will allow businesses to carry on operating in a "business as usual" manner [42]. Even if prices are still relatively low overall, carbon pricing internalises the cost of emissions and so helps to direct investments in low-carbon technologies. CCS is simply the process of trapping carbon dioxide that is released from power plants during the burning of oil, coal, or gas and sequestering it in unused geological reservoirs to keep it from entering the atmosphere. Therefore, CCS aims to prevent additional emissions from fossil fuels, but it should be noted that it is still a relatively new technology and is very speculative as a result of its slow adoption and lack of scaled-up implementation as well as uncertainty surrounding its effects on human health and the environment [43]. The two strategies are therefore dependent on one another; without a sufficiently high carbon price, CCS would continue to be economically undesirable.

The companies predict that these three strategies will give us time to transition to low-carbon technologies while still enabling us to use fossil fuels more sustainably for the better part of the next century. Although these proposals might be sometimes viewed as unattainable, they do represent an attempt on the part of the fossil fuel sector to reduce emissions, or rather, a bid to continue existing in the post-Paris, low-carbon world.

CONCLUSION

Hove *et al*. asserted in their study that the moral conundrum surrounding climate change that the oil industry faces can be summed up as a conflict between the economic need for a profitable oil industry and the fact that CO_2 emissions are causing life-threatening climatic changes. It is indisputable that lucrative fossil fuel companies produce CO_2 emissions as a by-product. The economy took a literal boost after fossil fuels made it into the market; the growth in developing countries caught pace and economies profited. Divestment strategies and other policies that attempt to move away from a world where fossil fuels are required have thus frequently begun to sound hazardous economically, which has slowed down efforts to mitigate climate change.

This conundrum has created divisions in the energy-intensive industry, with some placing a higher value on profits than emissions, others placing a higher value on emissions than profits, and yet others choosing to disregard the conundrum for the time being. Each of the three has its disadvantages and benefits. One cannot be certain of how the various oil, gas, and coal firms' earnings have changed in relation to measures to mitigate climate change. Due to a lack of data, it is difficult to determine whether a company's efforts to reduce emissions are being made out of genuine concern for the environment or are instead being made in order to increase profits or maintain their relevance in a world where divestment plans are accelerating and stranded assets are on the rise. It is difficult to tell which strategy is more ethical or generates greater profits due to the lack of statistics.

Richard Heede [44] discovered in his study that only 90 companies, of whom 56 are oil and gas firms, are responsible for 63% of historical CO_2 and methane emissions from 1751 to 2010. Heede refers to this group of 90 as the "carbon majors," and the majority of their emissions have taken place since 1986. Since only a small number of major oil corporations are the main contributors to greenhouse emissions, they have the greatest potential to bring about a positive change in the deteriorating climatic conditions and the environment. If we examine the sector's efforts to reduce its emissions, the future may appear bright, but sizable efforts are still required to mitigate the worst effects of climate change. The problem is that we cannot ignore the harm that fossil fuels are causing to the environment, the climate, and society, nor can we deny that the economy and we as individuals are addicted to them.

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